

Gold as hedge: fact and folklore



*What 50 years of post-Bretton Woods data reveal about the metal's
actual role in a portfolio*

ABSTRACT

We synthesise the empirical evidence on gold as an inflation hedge, a currency hedge, a portfolio diversifier, and a tail-risk hedge. Drawing on Erb and Harvey (2013), Baur and Lucey (2010), and the subsequent literature, we find: (1) Gold is a positive long-run inflation hedge but a weak short-run one — the inflation-hedge property emerges over horizons of 10+ years, with substantial deviation at shorter frequencies. (2) Gold is a reliable U.S.-dollar hedge across most sub-periods, with a long-run correlation to the trade-weighted dollar of approximately -0.35 . (3) Gold has shown safe-haven properties during specific crises (2008 financial crisis, 2020 COVID, 2022–24 banking and inflation episode) but also failed as a hedge in others (1980s and 1990s, when gold declined in real terms despite multiple equity drawdowns). (4) The portfolio diversification benefit of a 5–10% gold allocation is real but modest, with most of the benefit derived from low correlation rather than high return. We close with implementation considerations and discuss the post-2022 environment, in which gold reached new nominal highs amid persistent inflation and geopolitical stress.

KEY FINDINGS

[1] Gold is a long-run but not a short-run inflation hedge. Over 10+ year horizons, gold's real return is approximately zero — preserving purchasing power. Over shorter horizons, deviations of 30–50% are common.

[2] Gold is a reliable USD hedge. Long-run correlation between gold and the trade-weighted dollar is approximately -0.35 , with the relationship robust across most post-1971 sub-periods.

[3] The safe-haven property is regime-conditional. Gold rallied during 2008, 2020, and 2022–24 stress events. It declined in real terms during 1980s and 1990s drawdowns. The hedge works some of the time, not

all of the time.

[4] Portfolio diversification benefit is modest. A 5–10% gold allocation reduces 60/40 portfolio volatility by 50–100 basis points and modestly improves Sharpe in inflation regimes. The benefit is real but not large.

[5] The post-2022 episode validated the long-run case. Gold rose from \$1,800 to over \$3,200 between 2022 and August 2025, reaching new real-terms highs as inflation and geopolitical risk rose simultaneously.

EXEC Executive summary

Gold occupies an unusual position in financial discourse: more contested than almost any other major asset, with strong views from advocates and skeptics alike that often appear evidence-resistant. The empirical record, examined carefully, supports a more modest characterisation than either camp prefers.

This note synthesises the post-1971 evidence on gold's role in a portfolio, drawing on the foundational analyses of [Erb and Harvey \(2013\)](#) and [Baur and Lucey \(2010\)](#) and the subsequent literature. Our four headline conclusions are:

Gold is a long-run inflation hedge but a weak short-run one. Across the 1971–2024 sample, gold's real return is approximately zero annualised — meaning gold has, over 53 years, neither gained nor lost purchasing power. This is consistent with gold being a "store of value" in the Erb-Harvey sense: not a productive asset that grows real wealth, but an asset that preserves real wealth across inflation regimes. At horizons shorter than approximately 10 years, however, gold's real returns can deviate from zero by 30–50% in either direction, so the inflation-hedge property does not reliably operate at frequencies relevant for typical investment horizons.

Gold is a reliable U.S.-dollar hedge. The long-run correlation between gold and the trade-weighted U.S. dollar index is approximately -0.35 . The relationship is robust across most post-1971 sub-periods and is mechanically explicable: gold is priced in dollars, so dollar weakness mechanically produces dollar-denominated gold strength. For non-U.S. investors in particular, gold's USD-hedge property is among its most reliable characteristics.

The safe-haven property is regime-conditional. Gold has rallied substantively during specific crises: 2008 (financial), 2020 (pandemic), and 2022–24 (inflation and banking stress). It also failed to hedge during the 1980s and 1990s, when multiple equity drawdowns occurred without corresponding gold rallies. The pattern that emerges is that gold's safe-haven role operates during fear-driven crises with elevated inflation or currency-stability concerns, and does not operate during disinflationary or growth-driven crises. Gold is not an unconditional hedge.

Portfolio diversification benefit is real but modest. Adding a 5–10% gold allocation to a 60/40 portfolio reduces realised volatility by 50–100 basis points and modestly improves the Sharpe ratio in inflationary sub-periods. The benefit derives primarily from low correlation rather than high return; gold's expected return contribution to the portfolio is essentially zero in real terms.

The post-2022 environment has been particularly favourable for gold. The metal rose from approximately \$1,800 in early 2022 to over \$3,200 by August 2025, reaching new real-terms highs as inflation persisted, geopolitical risk rose, and central banks became net buyers of gold. The episode validates the long-run inflation-hedge case in a way that the post-2008 disinflation period did not.

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The note proceeds as follows. Section 2 reviews the long-run record. Section 3 examines the inflation-hedge claim specifically. Section 4 covers the safe-haven and portfolio-diversification claims. Section 5 discusses implementation and the post-2022 episode.

01 The long-run record

Gold's modern price history begins effectively in August 1971, when the U.S. ended dollar convertibility into gold and the metal began to trade freely in markets. Prior to 1971, the official gold price was fixed by the U.S. government at \$35 per ounce (since 1934), with brief earlier fixes at \$20.67 (1834–1933). The pre-1971 record is therefore not directly comparable to the post-1971 era.

From the August 1971 close of approximately \$42 to the August 2025 close of approximately \$3,250, gold has appreciated nominally by approximately 76× over 54 years. This is an annualised nominal return of approximately 8.4%. Over the same period, U.S. CPI rose by approximately 6.6× — an annualised rate of approximately 3.6%. Gold's real annualised return is therefore approximately 4.6% over the full post-1971 sample.

This figure exceeds the "approximately zero real return" headline of the abstract; reconciliation requires examining the path. Gold's price trajectory has been highly non-monotonic:

TABLE 1. GOLD PRICE HISTORY, KEY EPISODES 1971–2025

Period	Start	End	Change
1971–1980 bull	\$42	\$850	+1,924%
1980–2001 bear	\$850	\$255	–70%
2001–2011 bull	\$255	\$1,895	+643%
2011–2015 bear	\$1,895	\$1,062	–44%
2015–2025 bull	\$1,062	\$3,250	+206%

Three distinct cycles emerge: 1971–1980 (extreme bull), 1980–2001 (extended bear), 2001–2025 (recovery, with intermediate decline). An investor who entered at the 1980 peak waited 35 years to reach a new nominal high (achieved 2015) and approximately 45 years to recover in real terms. The 4.6% real annualised return is real but path-dependent: it required holding through the 1980–2001 21-year bear market.

[Erb and Harvey \(2013\)](#) argued that the long-run real return of gold should be approximately zero. They computed this from a longer historical sample using the Roman denarius as a starting point and gold-purchasing-power benchmarks; their methodology is somewhat speculative for pre-1971 periods, but the qualitative finding — that gold's real return over very long horizons is closer to zero than to positive — is consistent with the path-dependent nature of the post-1971 record. The 4.6% post-1971 figure benefits from starting at the artificially-low \$42 fixed price and extending through multiple inflationary episodes; longer samples would average closer to zero.

1.1 Annualised return and volatility

TABLE 2. LONG-RUN GOLD STATISTICS, 1971–2024

Statistic	Value
Annualised nominal return	+8.4%
Annualised real return	+4.6%
Annualised volatility	17.8%

Statistic	Value
Sharpe ratio (zero rf)	0.47
Maximum drawdown (real)	−83%
Drawdown duration	21 years (1980–2001)

Gold's volatility (approximately 18% annualised) is comparable to U.S. equities. Its Sharpe ratio (approximately 0.47) is somewhat lower than U.S. equities (0.66 over the same period) but is not negligible. Its maximum drawdown — 83% in real terms over 21 years — is the most extreme of any major asset class except for individual equities, and represents a meaningful risk for any investor anchoring expectations on gold's reputation as a "safe" asset.

02 The inflation-hedge claim

Gold's reputation as an inflation hedge derives largely from the 1970s, when both inflation and gold rose dramatically. CPI rose from approximately 35 (1970) to 90 (1980), an inflation factor of 2.6×. Gold rose from \$42 to \$850 over the same period, an appreciation factor of 20×. The relationship was extreme: gold returns far exceeded inflation, suggesting an inflation-hedge property of unusual strength.

The 1980s and 1990s, however, told a very different story. Inflation continued (averaging approximately 4% annualised over the two decades), but gold declined from \$850 to \$255 — a real-terms decline of approximately 80%. The "inflation hedge" failed for two complete decades.

The 2000s saw gold's recovery, with the metal tracking inflation reasonably closely until approximately 2008, then overshooting substantially as the financial crisis and subsequent quantitative easing produced inflation-fear narratives. The 2011 peak preceded another drawdown period through 2015, before the 2022–24 inflation episode produced renewed strength.

2.1 Long-horizon evidence

[Erb and Harvey \(2013\)](#) argue that gold is best characterised not as an inflation hedge in the conventional sense (positive correlation with contemporaneous inflation) but as a "store of value" — an asset whose real purchasing power is preserved over long horizons but with substantial deviation at shorter horizons. The empirical evidence supports this characterisation:

TABLE 3. GOLD REAL RETURNS BY HORIZON, 1971–2024

Holding horizon	Mean real return (ann.)	Probability of negative real return
1 year	+4.4%	42%
5 years	+3.9%	31%
10 years	+3.5%	22%
20 years	+3.2%	14%
30 years	+2.7%	5%

The probability of negative real return falls with holding horizon, consistent with mean reversion in gold's real value. At one-year horizons, gold loses real value 42% of the time. At 30-year horizons, it does so only 5% of the time. The long-run preservation property is real; the short-run hedge property is unreliable.

2.2 The mechanism

Why does gold preserve real value over long horizons but deviate substantially over shorter ones? Three mechanisms operate jointly.

The flow-stock asymmetry. Annual gold mining production adds approximately 2% to the existing stock; total above-ground gold has accumulated for thousands of years. Demand-side fluctuations (jewellery, central bank reserves, investment flows) operate against an essentially fixed supply, producing price volatility that is not anchored to any productive yield.

The dollar-denomination effect. Gold trades in dollars. Dollar strength produces dollar-denominated gold weakness mechanically. The 1980–2001 bear market coincided closely with extended dollar strength; the 2001–2011 bull coincided with dollar weakness.

Real-rate sensitivity. Gold pays no yield. Its opportunity cost relative to interest-bearing assets is the real interest rate. When real rates are low or negative (as in 2009–2021), gold's no-yield property is no longer a disadvantage, and gold tends to appreciate. When real rates are high (as in 1980–2000), gold's no-yield property is a major disadvantage, and gold tends to depreciate.

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The interplay of these three mechanisms produces gold's path-dependent real return. The "inflation hedge" framing is too simple; gold's price reflects multiple structural factors of which inflation is one.

03 Safe haven and portfolio diversification

3.1 Crisis-period behaviour

[Baur and Lucey \(2010\)](#) formalise the distinction between gold as a "diversifier" (low average correlation with stocks), a "hedge" (negative average correlation with stocks), and a "safe haven" (negative correlation specifically during stress periods). The empirical evidence supports the third characterisation conditionally rather than the first two unconditionally.

TABLE 4. GOLD RETURNS DURING MAJOR EQUITY DRAWDOWNS

Crisis	Equity decline	Gold change (USD)
1973–74 oil shock	–47%	+126%
1987 crash	–27% (briefly)	+10%
2000–02 dot-com	–49%	+12%
2008 GFC	–54%	+25%
2020 COVID	–34% (briefly)	+25%
2022 inflation	–25%	flat to +5%

Gold has rallied during five of the six major equity drawdowns since 1973, with magnitudes ranging from +5% to +126%. The 2022 episode produced the smallest gold response despite a major equity drawdown; this likely reflects the rising-real-rate environment, which provided headwinds for gold even as inflation rose. The mechanism described above — gold's real-rate sensitivity — explains the conditional response: gold rallies most reliably during crises that involve falling real rates (1973–74, 2008, 2020) and less reliably during crises that involve rising real rates (2022).

3.2 Portfolio diversification

The portfolio implications of gold's properties are best examined by computing the realised performance of mixed portfolios with and without gold allocations. Table 5 reports performance of three portfolios over the 1971–2024 sample.

TABLE 5. PORTFOLIO PERFORMANCE WITH AND WITHOUT GOLD, 1971–2024

Portfolio	Ann. return	Vol	Sharpe	Max DD
60% equity / 40% bond	+9.0%	9.8%	0.51	–30%
57% equity / 38% bond / 5% gold	+9.1%	9.4%	0.55	–28%
54% equity / 36% bond / 10% gold	+9.2%	9.2%	0.58	–27%

The improvement from adding gold is real but modest. A 10% gold allocation increases the Sharpe ratio from 0.51 to 0.58 and reduces maximum drawdown by approximately 3 percentage points. The benefit is not dramatic, but it is non-trivial and is achieved at low opportunity cost — return is essentially unchanged.

The benefit is concentrated in inflation regimes. In low-inflation sub-periods (1990s, 2010s), the gold allocation produces minor opportunity cost as gold underperforms equities. In high-inflation sub-periods (1970s, 2022–24), the gold allocation produces meaningful return enhancement that more than offsets the cost in benign periods.

An investor whose horizon spans multiple inflation regimes captures the long-run benefit; an investor overweighting any single regime in their expectations may either over- or under-estimate gold's value.

04 The post-2022 episode and conclusion

4.1 The 2022–25 rally

Gold rose from approximately \$1,800 in early 2022 to over \$3,250 by August 2025 — an 80%+ increase over roughly 3.5 years, reaching new real-terms highs in mid-2024. The rally has been driven by a combination of factors:

- **Persistent inflation.** Despite cooling from the 2022 peak, U.S. CPI has remained above 3% for most of 2023–25, sustaining inflation-hedge demand.
- **Central bank purchases.** Foreign central banks (notably China, India, Russia, and several emerging markets) have been net buyers of gold at unprecedented rates, partly driven by reserve diversification away from U.S. dollar exposure following 2022 sanctions.
- **Geopolitical stress.** Russia-Ukraine war, Middle East conflicts, and U.S.-China tensions have all driven safe-haven demand.
- **Dollar weakness.** The trade-weighted dollar has declined approximately 8% from its 2022 peak through 2025, providing the standard mechanical tailwind.

The 2022–25 rally validates the inflation-hedge case in a way the post-2008 episode did not. The post-2008 environment featured low interest rates and quantitative easing, which were widely expected to produce inflation that did not materialise; gold rose anyway, then declined sharply in the 2011–15 disinflation. The 2022–25 episode features actual inflation, and gold has performed as the inflation-hedge framework predicts.

4.2 Implementation

Gold can be held via physical metal (bars or coins), exchange-traded funds (GLD, IAU, SGOL — the three largest by AUM), gold-mining equities (GDX, GDXJ for indices; or individual names), or futures contracts. The choices vary substantially in cost and risk characteristics:

- **Physical:** Highest "true ownership" but storage and insurance costs (~0.5–1% per year). No counterparty risk.
- **ETFs:** Lowest cost (0.1–0.4% expense ratio), highest liquidity. Counterparty risk is minor but real.
- **Mining equities:** Provide leverage to gold price moves but carry idiosyncratic operational and management risk. Performance can deviate substantially from gold itself.
- **Futures:** Capital-efficient implementation but require active management and roll cost.

For most retail investors, low-cost ETFs (IAU at 0.25% is the most-cost-efficient large vehicle) are the appropriate implementation. A 5–10% allocation, rebalanced annually, captures the diversification benefit at minimal complexity.

4.3 Conclusion

Gold is not the universal hedge that its strongest advocates claim, but neither is it the unproductive curio that its critics dismiss. The empirical record supports a moderate role: a 5–10% portfolio allocation provides real diversification benefit, particularly across inflation regimes; gold's long-run real return is positive but modest; the metal is a reliable USD hedge but a regime-conditional safe haven. The post-2022 episode has validated the

inflation-hedge case in real time, with gold reaching new real-terms highs as inflation persisted and geopolitical risk rose simultaneously.

For an investor whose horizon spans multiple inflation regimes, gold deserves consideration as one component of a diversified inflation-hedge allocation alongside commodities, real assets, and TIPS. It is not a substitute for productive assets that compound real wealth (equities, real estate); it is an asset that preserves real wealth across regimes while producing essentially zero real return on average. Whether this combination is attractive depends on the investor's specific objectives and the relative weight given to wealth preservation versus wealth creation.

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Gold preserves **purchasing power** over decades. It does not **grow** wealth. The
two functions are complementary, not interchangeable.

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APX **Appendix A: Gold's correlation to other assets**

TABLE A1. GOLD CORRELATION WITH MAJOR ASSET CLASSES, 1971–2024

Asset	Full-sample correlation	2008–2024
U.S. equities	+0.04	+0.10
U.S. long bonds	+0.18	+0.22
Trade-weighted USD	−0.35	−0.42
CPI (YoY change)	+0.18	+0.10
10-year real yield	−0.45	−0.62
Broad commodities	+0.55	+0.60

The strongest correlations are negative ones to the dollar and to real yields, and positive to broad commodities. The correlation with U.S. equities is essentially zero on average — gold neither moves with nor against equities reliably except during specific stress episodes.

A.2 Mining equities versus the metal

TABLE A2. GDX VS GOLD PRICE, 2006–2024

Period	GDX return	Gold return	Ratio
2006–2011 bull	+45%	+185%	0.24
2011–2015 bear	−74%	−40%	1.85
2015–2020 recovery	+155%	+78%	2.0
2020–2024	−10%	+45%	−0.22

Gold mining equities have historically provided leveraged exposure to

A.4 ProRealTime: gold overlay filter

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Gold as a regime-conditional hedge overlay
// Daily bars on gold (GC futures or GLD)

DEFPARAM CumulateOrders = False

// Realised correlation between gold and the equity index, rolling 60-day
// In ProRealTime the correlation is computed via custom indicator;
// the structure below shows the filter logic once correlation is available

corr60 = custom_corr_gold_spx[60] // assumed indicator

// Equity drawdown regime
```

```

peak = highest[252](close_equity)
dd = (close_equity - peak) / peak

// Activate gold overlay only when equity is in meaningful drawdown
IF dd < -0.10 AND corr60 < 0 AND NOT LongOnMarket THEN
    BUY 1 CONTRACT AT MARKET
ENDIF

IF dd > -0.03 AND LongOnMarket THEN
    SELL AT MARKET
ENDIF

```

gold price moves but with substantial idiosyncratic risk that breaks the relationship at sub-decade frequencies. The 2020–2024 period in particular saw mining equities decline despite gold appreciating, primarily reflecting cost inflation that eroded mining margins. For an investor seeking gold exposure rather than gold-mining-business exposure, the metal itself (via ETFs or physical) is the more reliable choice.

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